

Alan Luu

Software Engineer – Backend & Data Platform

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EXPERIENCE

Azure CXP Software Engineer (Data/Backend)

Apr 2024 – Dec 2025

Microsoft

Redmond, WA

- Migrated legacy Scala JAR logic to PySpark in Azure Synapse, rewriting 20+ notebooks and data pipelines to improve scalability and maintainability of Fortune 500 consumption and billing analytics (e.g., Adobe, NVIDIA).
- Reduced runtime of 3 production Apache Spark data pipelines by 75% (24h → 5–6h) by optimizing joins and partitioning, resolving stage retries through cluster scaling, and updating authentication to Managed Identity.
- Built ETL pipelines using Cosmos DB, Cosmos SCOPE, and Kusto/KQL to ingest, validate, and onboard 80+ metrics to an Azure Usage dashboard, deploying JSON configs to render graphs for telemetry and cloud metrics.
- Owned and enhanced 150+ Playwright E2E tests (TypeScript), expanding coverage across Product, Consumption, and Quality domains while stabilizing flaky tests and validating Azure Usage dashboard features amid UI changes.

Xbox Software Engineer (Systems)

Mar 2022 – Sep 2023

Microsoft

Redmond, WA

- Developed and maintained multi-day (1–12 day) reliability test suites (C#, C++) for next-gen Xbox consoles, validating hardware, OS, networking, and thermal performance, ensuring timely delivery to global testing teams.
- Stabilized long-running Xbox reliability tests by debugging and resolving memory leaks, port exhaustion, and cross-stack failures across firmware, OS, networking, and test infrastructure, unblocking global teams.
- Improved code quality and security across 4 production repositories by resolving 120+ CodeQL issues (C#, C++), addressing vulnerabilities and maintainability risks.

Xbox Software Engineer Intern (Full Stack)

Feb 2016 – Aug 2016

Microsoft

Redmond, WA

- Developed Cosmos SCOPE analytics scripts to extract and summarize Cortana speech feature usage, helping developers identify high-impact features for the Xbox One Cortana release.
- Built OnePulse dashboard metric tiles (C#) to visualize Cortana usage trends, enabling engineering teams to prioritize feature improvements.
- Implemented an ASP.NET web dashboard aggregating Cortana metrics, improving access to usage insights.

PROJECTS

RetailLens | [GitHub](#) | [Live](#) | AWS, S3, Glue, PySpark, Athena, DynamoDB, FastAPI, React

June 2026

- Built a full-stack AWS retail analytics platform that ingests raw CSV data into S3, catalogs it with Glue, transforms it into partitioned Parquet using Glue PySpark, computes aggregate metrics with Athena, and serves precomputed analytics from DynamoDB through FastAPI and React.

Job Processing API | [GitHub](#) | [Live](#) | Python, FastAPI, SQLAlchemy, pytest, Docker

May 2026

- Built a job-processing backend for creating, claiming, and processing jobs through guarded state transitions, with SQLite persistence, an in-memory priority queue, pytest coverage, and Docker support.

EDUCATION

Bachelor of Science in Computer Science

Sep 2018 – Aug 2020

University of Washington

Bothell, WA

Associate in Arts and Science

Jan 2014 – Mar 2018

Bellevue College

Bellevue, WA

Apprenticeship in Data Analyst

Jun 2015 – Aug 2016

Year Up

Redmond, WA

TECHNICAL SKILLS

Languages: Python, SQL, C#, Java, JavaScript/TypeScript, C++, Scala

Cloud & Data: Azure, Azure Synapse, Azure Data Factory, Apache Spark/PySpark, ETL/ELT, Cosmos DB, Cosmos SCOPE, Kusto/KQL, Azure Blob Storage, SQL Server, PostgreSQL, MySQL, AWS, S3, Glue, Athena, DynamoDB, EC2

Frameworks & Libraries: FastAPI, SQLAlchemy, Pydantic, ASP.NET, React, Node.js, Next.js, Spring Boot

Developer Tools & Platforms: Azure DevOps (CI/CD), Git, pytest, Playwright, CodeQL, Docker, Linux, VS Code